

# ARJUN VASHISHTHA

Full-Stack Developer • AI Builder • Data Science • SDN Engineer

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## PROFESSIONAL SUMMARY

Fourth-year B.Tech CSE student at VIT Bhopal, currently leading Website Management & Marketing at AIOorders × Foodswipe. Bridges product, data, and growth — managing websites across two brands, coordinating software delivery, and driving marketing through UGC content and paid social campaigns. Strong foundation in Python, machine learning, and data analytics, with hands-on experience building autonomous AI agents, full-stack web applications, SDN controllers, and data-driven solutions. Passionate about shipping practical products that solve real problems. Live portfolio: [arjun-portfolio-emc.pages.dev](https://arjun-portfolio-emc.pages.dev).

## EXPERIENCE

### Website Management & Marketing Head

AIOorders × Foodswipe | 2024 – Present | Remote

- Leading website management and marketing across both AIOorders and Foodswipe brands.
- Manage website updates, performance tuning, and content drops for two products simultaneously.
- Plan and run marketing campaigns across Instagram, Facebook, and Google Ads to drive reach and engagement.
- Create UGC content and produce/edit video to support brand and product marketing initiatives.
- Coordinate software delivery timelines and own cross-functional comms between dev and growth teams.

## KEY PROJECTS

### 🔗 PORTFOLIO HIGHLIGHT

#### Dungeon RPG Portfolio — [arjun-portfolio-emc.pages.dev](https://arjun-portfolio-emc.pages.dev)

Premium dungeon/RPG-themed interactive portfolio built with Next.js 16, Framer Motion, and Web Audio API. Features 8-layer parallax cave backgrounds, animated King character with click-triggered attack sequences, Goddess NPC with 132 sarcastic dialog lines and 40+ topic knowledge base, achievement system (10 unlockables), treasure chest loot, RPG-style project cards with rarity borders, ambient castle music, and a dedicated Experience page with an interactive wizard character. Deployed on Cloudflare Pages.

14+ projects built across AI/ML, computer vision, networking, full-stack, frontend, and backend. Live demos and source code available on GitHub. Selected highlights:

#### SDN Controller — Ryu + P4 + DDoS Detection

Custom Software-Defined Networking controller built on Ryu + OpenFlow 1.3 with a P4-16 data plane for BMv2. Custom LLDP topology discovery, reactive L2 forwarding with BFS shortest-path computation, and a real-time DDoS detector using threshold + entropy anomaly detection that auto-installs ACL drop flows on every switch. Includes a Flask REST API (16 endpoints), D3.js real-time topology dashboard, sdnctl CLI tool, and a 66-test pytest suite with GitHub Actions CI on Python 3.9/3.10/3.11.

Tech: Python, Ryu, P4-16, OpenFlow 1.3, Mininet, Flask, D3.js, pytest, GitHub Actions

Link: [github.com/arjundroid12/sdn-project](https://github.com/arjundroid12/sdn-project)

#### SpellCaster — Gesture-Controlled Spellcasting

Gesture-controlled spellcasting web app. Make a hand sign with your webcam — MediaPipe Hands detects 21 landmarks per hand in real time, the gesture classifier smooths out jitter, and a spell fires from your fingertip with neon particle effects + synthesized Web Audio sounds. Six spells: Fireball (fist), Shield (open palm), Lightning (V sign), Magic Missile (pinch), Heal (thumbs up), Ice Blast (flat sideways). Each spell has its own cooldown, particle system (burst/beam/ring/trails), and synth sound. Pure vanilla JS — no build step, no deps, runs entirely in the browser. Live demo deployed on GitHub Pages.

Tech: MediaPipe Hands, Web Audio API, Canvas 2D, Vanilla JS, GitHub Pages

Link: [arjundroid12.github.io/spellcaster](https://arjundroid12.github.io/spellcaster)

#### SmartAgro — AI Plant Disease Detection

AI system that helps farmers detect plant diseases and get remedies from image or text input. Uses CNN/Random Forest for image classification and NLP for remedy queries, with a web interface and recent-query history.

Tech: Python, CNN, Random Forest, NLP, Web Interface

Link: [github.com/arjundroid12/SmartAgro-A-disease-detection-model-with-Human-Interaction](https://github.com/arjundroid12/SmartAgro-A-disease-detection-model-with-Human-Interaction)

### FIOLA — AI Voice Assistant

Custom voice assistant built with LLMs and Python for speech recognition and task automation. Handles natural language commands and executes tasks autonomously.

Tech: Python, LLM, Speech Recognition

### AI Agents Suite (4 Agents)

Built four autonomous AI agents: (1) Research Agent using the ReAct pattern for autonomous reasoning, (2) Multi-Agent System where 3 AIs collaborate to solve problems, (3) Data Analyst Agent that writes and executes Python in-browser via Pyodide, (4) Coding Agent — a mini v0.dev that generates HTML/CSS/JS from natural language descriptions.

Tech: Next.js, Cerebras, Pyodide, ReAct Pattern, Multi-Agent Systems

### Gesture Particle Painter & Gesture Volume Mixer

Two MediaPipe Hands projects: a gesture-controlled particle painter (paint with hand movements, mouse fallback) and a volume mixer controlled by hand height, pinch-to-mute, and two-hand dual-channel control.

Tech: MediaPipe Hands, Web Audio API, Canvas API, JavaScript

### Realtime Chat & Realtime Whiteboard

Multi-user real-time applications built with Socket.io: a multi-room chat with typing indicators and online user list, and a collaborative whiteboard with shared canvas, colors, brush sizes, and undo functionality.

Tech: Node.js, Express, Socket.io, WebSockets, Canvas API

Link: [github.com/arjundroid12/realtime-chat](https://github.com/arjundroid12/realtime-chat)

### JWT Auth Demo & URL Shortener API

Production-style backend projects: JWT authentication with bcrypt password hashing, refresh tokens, and token rotation; REST API for URL shortening with click analytics, custom aliases, and expiry support.

Tech: Node.js, Express, JWT, bcrypt, LowDB

Link: [github.com/arjundroid12/jwt-auth-demo](https://github.com/arjundroid12/jwt-auth-demo)

### Frontend App Suite (6 Apps)

Six vanilla JS applications with live demos: Calculator (custom expression parser), Notes App (custom markdown parser, tags, search), Weather App (Open-Meteo API, 5-day forecast), Kanban Todo (drag & drop), Movie Explorer (search, filters, favorites), Pomodoro Timer (stats charts, notifications). All deployed on surge.sh.

Tech: Vanilla JavaScript, HTML5, CSS3, localStorage, Chart.js, various APIs

Link: [github.com/arjundroid12](https://github.com/arjundroid12) (see repos)

## CORE SKILLS

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**Data Science & Machine Learning:** Python (Pandas, NumPy, scikit-learn), CNN, Random Forest, NLP, exploratory data analysis, model building & evaluation

**AI Engineering:** Cerebras, OpenAI API, ReAct Pattern, Multi-Agent Systems, Pyodide, LLM integration

**Computer Vision:** MediaPipe Hands, hand pose tracking, gesture recognition, real-time webcam processing, Canvas 2D rendering

**Networking:** Software-Defined Networking (SDN), Ryu Controller, P4, OpenFlow 1.3, Mininet, network topology discovery, DDoS detection

**Web Development:** JavaScript, TypeScript, React, Next.js 16, Node.js, Express, Tailwind CSS, Framer Motion, Three.js

**Data & Analytics:** Power BI, MySQL, Excel, data cleaning & preprocessing, data visualization

**Software & Product:** Website management, project coordination, AI website building, no-code app development, REST API design

**Marketing & Content:** UGC content creation, social media advertising (Instagram, Facebook, Google Ads), video editing & videography

**Tools & Platforms:** Git, GitHub Actions, Vercel, Render, Cloudflare Pages, Three.js, MediaPipe, Web Audio API, Flask

## EDUCATION

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### B.Tech in Computer Science & Engineering

VIT Bhopal University, Bhopal | 2022 – 2026 | Fourth Year (Undergraduate)

## INTERESTS

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Music (vocalist & flutist), videography, content creation, social media, and game development. Active GitHub contributor with daily coding challenges auto-committed via GitHub Actions.